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Special Section:

Machine learning for Solid Earth observation, modeling and understanding

Key Points:

- Physics-informed neural network (PINN) can efficiently solve full waveform seismic inversions in 2D acoustic media with a rather simple and straightforward implementation
- PINN can seamlessly handle physical constraints and absorbing boundary conditions relevant to geophysical applications
- PINN provides a robust framework to encode a priori knowledge of the subsurface and the inclusion of multiple data type in joint inversions

Supporting Information:

Supporting Information may be found in the online version of this article.

Correspondence to:

M. Rasht-Behesht,
seyed_hossein_rashtbehesht@brown.edu

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Physics-Informed Neural Networks (PINNs) for Wave Propagation and Full Waveform Inversions

Majid Rasht-Behesht¹ , Christian Huber¹ , Khemraj Shukla², and George Em Karniadakis³

¹Department of Earth Environmental and Planetary Sciences, Brown University, Providence, RI, USA, ²Division of Applied Mathematics, Brown University, Providence, RI, USA, ³Division of Applied Mathematics and School of Engineering, Brown University, Providence, RI, USA

Abstract We propose a new approach to the solution of the wave propagation and full waveform inversions (FWIs) based on a recent advance in deep learning called physics-informed neural networks (PINNs). In this study, we present an algorithm for PINNs applied to the acoustic wave equation and test the method with both forward models and FWI case studies. These synthetic case studies are designed to explore the ability of PINNs to handle varying degrees of structural complexity using both teleseismic plane waves and seismic point sources. PINNs' meshless formalism allows for a flexible implementation of the wave equation and different types of boundary conditions. For instance, our models demonstrate that PINN automatically satisfies absorbing boundary conditions, a serious computational challenge for common wave propagation solvers. Furthermore, a priori knowledge of the subsurface structure can be seamlessly encoded in PINNs' formulation. We find that the current state-of-the-art PINNs provide good results for the forward model, even though spectral element or finite difference methods are more efficient and accurate. More importantly, our results demonstrate that PINNs yield excellent results for inversions on all cases considered and with limited computational complexity. We discuss the current limitations of the method with complex velocity models as well as strategies to overcome these challenges. Using PINNs as a geophysical inversion solver offers exciting perspectives, not only for the full waveform seismic inversions, but also when dealing with other geophysical datasets (e.g., MT, gravity) as well as joint inversions because of its robust framework and simple implementation.

Plain Language Summary Recent advancement in machine learning have provided new paradigms for scientists and engineers to solve challenging problems. Here we apply a new strategy in machine learning (physics-informed neural networks (PINNs)) to seismic imaging, that takes advantage of the governing physical laws to complement the limited data available (seismograms). We design synthetic seismic imaging experiments and demonstrate that PINNs provides a computationally simple and efficient alternative to traditional seismic inversion solvers. The method proposed in this study can be naturally extended to other geophysical data inversions based on a similar framework.

1. Introduction

Seismic inversions are important tools for imaging heterogeneities in the Earth's subsurface. However, the resolving power of seismic inversion techniques is strongly controlled by the type and extent of the data used from seismograms. Generally, full waveform inversions (FWIs) can provide significantly better results than travel-time tomography, particularly for scenarios where the dominant wavelength of the seismic wave is greater than the length-scale of the target heterogeneities (Cassidy, 1992; Fichtner & Trampert, 2011; Marquering et al., 1999; Rasht-Behesht et al., 2020). Regardless of their advantages, FWIs remain technically and computationally challenging. Adjoint methods (Bozdağ et al., 2016; Fichtner et al., 2006; Plessix, 2006; Tromp et al., 2008) offer an efficient strategy to reduce computational cost for FWIs from N forward simulations per optimization step (N = number of model parameters) to two forward simulations. However, the derivation and implementation of adjoint methods for FWIs remains challenging and must be treated on a case-by-case basis for different systems (Bozdağ et al., 2016; Cockett et al., 2015).

With the successful application of machine learning, and in particular deep learning techniques, and the concurrent explosion of recorded seismic data made available over the past decade, seismologists have begun to search for modern efficient techniques (Bergen et al., 2019; Kong et al., 2019) to tackle problems such as earthquake detection, automatic phase picking (Mousavi et al., 2020; Ross et al., 2018; Yoon et al., 2015) and seismic

signal denoising (Zhu et al., 2019). A common theme thus far in the majority of these applications has been the abundance of labeled/unlabeled training data sets. As a result, areas such as seismic imaging applications where one is limited to spatially sparse data sets (seismometers), have not benefited as much from these methodologies. However, under conditions where a computationally inexpensive forward model of the wave propagation exists, such as when considering a 1D layered Earth structure, it is possible to generate large synthetic data sets to train a deep neural network (DNN) that can be then used as an efficient forward model surrogate (Araya-Polo et al., 2018; Moseley, Markham, & Nissen-Meyer, 2020; Moseley, Nissen-Meyer, & Markham, 2020). The intent of this approach is to make the solution to the forward problem more cost effective than the numerical solution of the wave equation and thus accelerate the procedure for inverse problems in seismology. The drawback of this technique is that it heavily relies on preexisting fast forward solvers for training purposes, but once trained, the neural network (NN) can be used as an independent machinery. Another approach, to accelerate the calculations of the derivatives in gradient-based techniques such as adjoint-methods is the use of reverse-mode automatic differentiation efficiently implemented in current deep learning libraries such as TensorFlow (Abadi et al., 2016) and PyTorch (Paszke et al., 2017) to solve the adjoint state FWIs (Zhu et al., 2021).

Raissi et al., 2019 have introduced a new class of DNNs known as physics-informed neural networks (PINNs) that provide an improved framework to overcome the obstacles pertaining to the training of DNNs caused by challenges in training data acquisition. PINNs take advantage of the governing physical laws behind the processes that generate the data, hence reducing greatly the reliance of the DNN on the labeled data during the training process. PINNs simply bridge the gap between data-scarcity and the data-intensive nature of DNNs and have been successfully applied to various problems in engineering and biology with applications including but not limited to heat transfer (S. Cai, Mao, et al., 2021; S. Cai, Wang, et al., 2021), solution of Navier-Stokes equations in fluid mechanics (Wu et al., 2018; Jin et al., 2021), high speed fluid flow (Mao et al., 2020) and solid mechanics (Haghighat et al., 2021). For a recent review of PINNs application see Karniadakis et al., 2021.

Haghighat et al., 2021 explored PINNs' application to linear elasticity and nonlinear plasticity and showed that they can be efficiently applied to inversion and surrogate modeling in solid mechanics. Shukla et al., 2020; Shukla, Jagtap, & Blackshire, 2021; Shukla, Jagtap, & Karniadakis, 2021 used PINNs for a nondestructive quantification of surface cracks and identification of microstructural properties of polycrystalline Nickel using ultrasound data. Moseley, Markham, & Nissen-Meyer, 2020; Moseley, Nissen-Meyer, & Markham, 2020, used PINNs as a solver for the forward acoustic wave propagation, while Smith et al., 2020 and Waheed et al., 2021 applied PINNs to the Eikonal equation as a forward solver for first arrival-time prediction and travel time tomography, respectively. Song et al., 2021 solved the frequency-domain anisotropic acoustic wave equation with PINNs. Nevertheless, to our knowledge, we present the first FWI for seismological applications using PINNs. In this study, we focus on the development of acoustic FWI with PINNs and demonstrate its practical application to various synthetic case studies. The salient results of our study are:

1. In most applications of PINNs, authors have incorporated training data sets from within the computational domain from other solvers or experimental data, which greatly facilitates the training process. In contrast, with seismic inversions, this is generally not possible (records of the wavefield are generally limited spatially to the surface or boreholes). We show that this limitation does not prevent PINNs from performing efficient and accurate seismic inversions.
2. We discuss the implementation of various types of boundary data, including a stress-free constraint on the top surface of the physical domain and absorbing boundary layers necessary for the simulation of wave propagation in infinite media. We show that the implementation of these constraints can be seamlessly handled with PINNs.
3. We present specific normalization guidelines that are crucial to PINNs' convergence as applied to the wave equation.
4. We demonstrate how to handle multiple seismic sources to improve the illumination of complex structural heterogeneity at depth.

This manuscript provides a first proof-of-concept that PINN models can be applied efficiently to seismic inversions within heterogeneous media. We also identify and discuss potential challenges in formulating a generalized approach to deal with complex structures. As deep learning and, in particular PINN methods, are rapidly improving, we discuss future direction to overcome these challenges.

2. Methods

2.1. Acoustic Wave Propagation

The propagation of acoustic waves in a 2D medium with negligible density variations and in the absence of body forces can be described in terms of the scalar wave potential ϕ as:

$$\alpha^2 \nabla^2 \phi + f = \frac{\partial^2 \phi}{\partial t^2}, \quad (1)$$

where $\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial z^2}$ is the Laplacian operator defined in the cartesian coordinate system, f is the external surface forces and α characterizes the acoustic wavespeed that strictly depends on the material properties of the medium through the relation $\alpha = \sqrt{\frac{K}{\rho}}$ with K and ρ being the material bulk modulus and density, respectively. Without loss of generality, we set $f \equiv 0$ and instead, enforce external forces through a perturbation of the initial field acting at some early time. The displacement field is retrieved from the gradient of the wave potential that is, $(u_x, u_z) = \nabla \phi$.

The forward simulation of the wave propagation involves solving Equation 1 given a set of boundary constraints and two early-time snapshots of the wave propagation as well as a precise knowledge of the material properties (wavespeed α) of the medium in the computational domain. Note that the first time-snapshot constrains the shape of the seismic source whereas the second enforces its propagation direction. In the inverse problem, α is either unknown or there exists some a priori information and it is evaluated spatially from the recorded ground motion. In this study we will investigate how PINNs can be used to solve these two classes of problems.

2.2. Neural Network (NN)

PINNs' architecture has been almost exclusively based on fully connected feed-forward NN. In the absence of any justifiable reasons to do otherwise, we, therefore, define a fully connected feed-forward NN with an input layer consisting of the physical coordinates x, z and time t , L hidden layers and an output layer representing the scalar acoustic wave potential $\phi \in \mathbb{R}$ (Figure 1). The various other physical variables, such as displacement or pressure, are obtained through the *automatic differentiation* of the wave potential NN using TensorFlow (Abadi et al., 2016). Note that the proper choice of the independent physical variable(s) that the output of the NN represents is problem-dependent. For instance, Haghighat et al., 2021 selected a separate NN for each component of the displacement field for problems in elasticity.

There are N_l number of neurons in the l th hidden layer. A linear transformation followed by a nonlinear neuron-wise activation function (σ) is applied at the input $x^{l-1} \in \mathbb{R}^{N_{l-1}}$ to the l th layer:

$$H(x^{l-1}) = \sigma(w^l x^{l-1} + b^l),$$

where $w^l \in \mathbb{R}^{N_l \times N_{l-1}}$ is the matrix of weights and $b^l \in \mathbb{R}^{N_l}$ is the vector of biases corresponding to the l th layer. The successive operation of this transformation law results in the final output of the NN with a total number of $\sum_1^L (N_l \times N_{l-1}) + b^l$ tunable parameters ($N_0 = 3$). We choose $\sigma = \tanh(\cdot)$ or $\sin(\cdot)$ as the nonlinear activation function for all NNs used in this study. The activation function acting on the last hidden layer to yield the output is the identity function. The interested reader is directed to Jagtap et al., 2020a, 2020b for a discussion of adaptive activation functions and their effect on convergence. The network's parameters are initialized from *independent and identically distributed (iid)* samples.

2.3. PINNs for the Forward Problem

We follow PINNs' original framework (Raissi et al., 2019), to obtain the parameters of a NN such that it closely approximates the acoustic wave potential ϕ . To do so, we enforce this NN to satisfy the partial differential Equation 1 with corresponding early-time snapshots of the wave propagation and boundary data computed on a set of randomly selected training data points (Figure 2). In the following, we define the various residual terms that we aim to minimize:

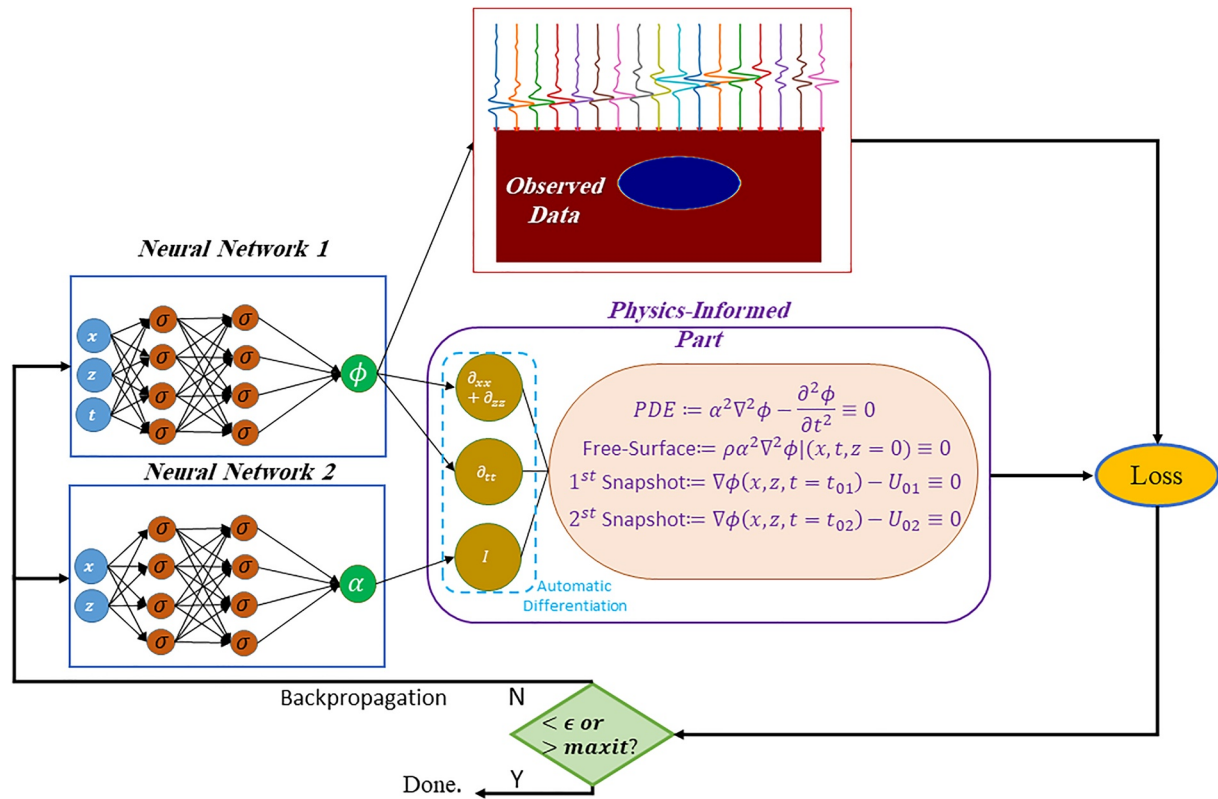


Figure 1. Schematics of physics-informed neural networks' (PINNs) workflow. Left: Fully connected feed-forward neural network, the output of which approximates the solution to the forward and inverse problems. Right: The governing physical laws and the observed real-world data, that is, seismograms, used to optimize the parameters of the PINN. The training stops when the loss error becomes smaller than a threshold, or the number of iterations goes beyond a set value.

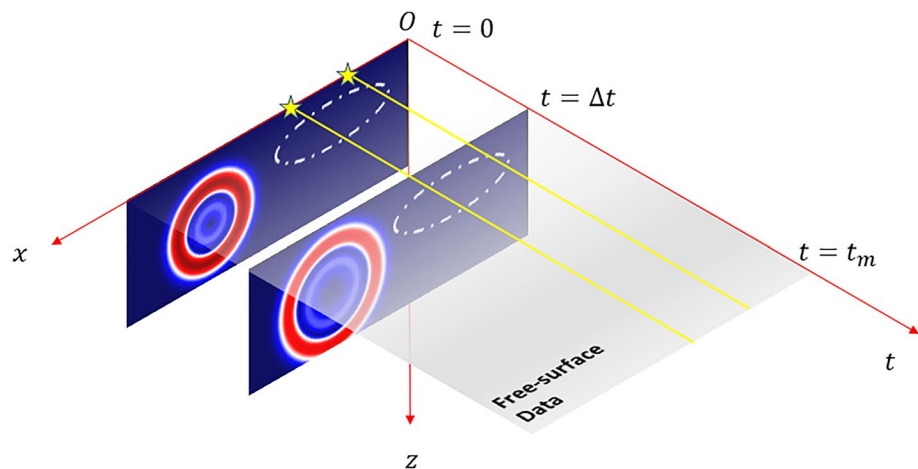


Figure 2. Schematic representation of a hypothetical computational domain (x , z , and t) with physics-informed neural networks. The two time-snapshots at times 0 and Δt are the only labeled data used from within the computational domain. The time-snapshots are color-coded for displacement amplitude. The white dashed line encloses a hypothetical heterogeneity. The gray hyperplane represents domain where the training data to apply, for example, a free-surface condition at the top of the domain. The two yellow stars represent the position of seismometers. t_m is the duration of the time domain. The PDE training data is selected randomly from the entire computational domain. Note that for some of the case studies we only use a subset of the various data sets illustrated here.

$$\begin{aligned}
 R_{\text{PDE}} &:= \alpha^2 \nabla^2 \phi - \frac{\partial^2 \phi}{\partial t^2} && \text{PDE} \\
 R_{P.C} &:= \rho \alpha^2 \nabla^2 \phi(x, z, t = 0) && \text{Free-Surface Constraint} \\
 R_{S_1} &:= \nabla \phi(x, z, t = t_1^0) - \overline{U_1^0}(x, z) && \text{First time-snapshot} \\
 R_{S_2} &:= \nabla \phi(x, z, t = t_2^0) - \overline{U_2^0}(x, z) && \text{Second time-snapshot} \\
 R_{\text{obs}} &:= \nabla \phi(x, z, t) - \overline{U_{\text{obs}}}(x, z, t) && \text{Observed data (For inversions)}
 \end{aligned}$$

$R_{P.C}$ indicates the physical free-surface constraint at the top of the model where pressure is fixed to zero. Observed data refers to the recorded wavefield on a sparse set of receivers (seismograms). In this study, the observed data in the form of synthetic seismograms and the early-time snapshots are obtained from SpecFem2D simulations (Komatitsch & Tromp, 1999; Tromp et al., 2008), a spectral element model for solving the wave equations in elastic/acoustic media. For the sake of convenience, we choose the time-snapshot data in terms of displacement, however one could choose one of these to explicitly involve the velocity field as well. Alternatively, one could also pose the training problem as an initial value problem, where the wavefield snapshots for training are provided in the form of two initial conditions at $t = 0$. It is important to note that modeling the wave propagation forward with only two early time snapshots of the wavefield cannot be implemented with traditional solvers, which illustrates the flexibility of PINNs in dealing with data at any time when applied to the wave equation.

The objective of the training process is to minimize the sum of mean squared errors:

$$\text{MSE}(\Theta) = \lambda_1 \text{MSE}_{\text{PDE}} + \lambda_2 \text{MSE}_S + \lambda_3 \text{MSE}_{P.C} + \lambda_4 \text{MSE}_{\text{Obs}} \quad (2)$$

Where $\Theta = W \cup b$ is the union of all the weights and biases of the NN. $\text{MSE}_{\text{PDE}} = \frac{1}{N_{\text{PDE}}} \sum_{i=1}^{N_{\text{PDE}}} |R_{\text{PDE}}(x_i, z_i, t_i)|^2$ is the loss term corresponding to the wave equation evaluated on a set of N_{PDE} randomly chosen PDE training data $(x_i, z_i, t_i) \in \Omega$ with $\Omega = \mathbb{R}^2 \times \mathbb{R}$ and

$$\text{MSE}_S = \frac{1}{N_{S_1}} \sum_{i=1}^{N_{S_1}} |R_{S_1}(x_i, z_i, t_i = t_1^0)|^2 + \frac{1}{N_{S_2}} \sum_{i=1}^{N_{S_2}} |R_{S_2}(x_i, z_i, t_i = t_2^0)|^2$$

represents the loss terms corresponding to the two vectorial early-time snapshot data $\overline{U_1^0}$ and $\overline{U_2^0}$ in terms of displacement. Similarly, the free-surface constraint and the observed data loss terms are defined as, $\text{MSE}_{P.C} = \frac{1}{N_{P.C}} \sum_{i=1}^{N_{P.C}} |R_{P.C}(x_i, z_i, t_i)|^2$ and $\text{MSE}_{\text{Obs}} = \frac{1}{N_{\text{obs}}} \sum_{i=1}^{N_{\text{obs}}} |R_{\text{obs}}(x_i, z_i, t_i)|^2$, respectively.

In contrast with standard partial differential equations solvers applied to the wave equation, PINNs essentially cast the forward (and inverse) problems as an optimization with optimal weights and biases Θ^* obtained by minimizing Equation 2, that is, $\Theta^* = \text{argmin}\{\text{MSE}(\Theta)\}$.

The hyperparameters $\lambda_{i,s} > 0$ in Equation 2 are set to normalize the different loss terms to guarantee a convergence to the correct solution. Failing to select proportionate loss terms would result in delayed convergence or possibly convergence to the wrong solution. Following the common practice in PINNs' literature, we find the proper values of $\lambda_{i,s}$ heuristically from trial and error; however, a more dynamic updating of these weights throughout the training process could be beneficial particularly for problems with persisting large residual errors at the boundaries (B. Wang et al., 2020; S. Wang, Teng, & Perdikaris, 2020; S. Wang, Yu, & Perdikaris, 2020). The learning rate annealing algorithm that Wang et al. proposed utilizes gradient statistics during the training process that would help maintaining a balance between different loss terms in Equation 2.

2.4. PINNs for the Seismic Inverse Problem

The goal of the inverse problem is to estimate the spatial distribution of the wavespeed α in Equation 1 from the data collected at the surface or in a borehole by a set of seismometers. We use an additional NN with independent weights and biases to estimate the distribution of seismic wavespeed in the medium (Haghighat et al., 2021; Shukla et al., 2020; Tartakovsky et al., 2020). This secondary network is generally significantly smaller in size (depth and

Table 1
Parameters Used for the Simulations

Type	Inhomogeneity type	Source type & frequency	Network [Width]*Depth	Activation function	N_{pde}	N_s	N_{PC}	$N_{Seismograms}$
Forward	Gaussian Anomaly	1 Point source 2 Hz	[50]*4	Sin()	20,000	2 *3600	N/A	N/A
Inverse	Homogeneous	1 Point source 20 Hz	[50]*4	tanh()	10,000	2 *3600	5000	2*1620
Inverse	Ellipsoidal Anomaly	1 Point source 20 Hz	[100]*8	tanh()	40,000	2 *3600	5000	2*1620
Inverse	Ellipsoidal Anomaly	2 Plane waves 2 Hz	[100]*8	tanh()	60,000	2 *3600	5000	2*1700
Inverse	Sinusoidal Checkerboard	9 Point sources 2 Hz	[100]*10	Sin()	60,000	2 *3600	N/A	2*5800

Note. [50]*4 indicates 4 hidden layers and 50 neurons in each. Note that the Neural Networks (NNs) in this Table indicate the forward NN. We use the same wavespeed NN for all the case studies here with 4 layers and 20 neurons in each. $N_{Seismograms}$ is the total number of training data from all the input seismograms in both x and z directions. All the input data sizes denote the batch size for the corresponding loss term.

width of the network) than the primary network estimating the wavefield, since the structural complexity is much simpler in nature than the wavefield's variations in space and the distribution of wavespeed does not depend on time. We choose a fixed wavespeed NN architecture for all the inverse case studies in this paper, namely, a fully connected feed-forward NN with 5 hidden layers and 20 neurons per layer. An important outcome of the PINNs' formalism is that it eliminates the need for a user-defined parameterization of the computational domain for the inverse problem and the related biases imposed from such parametrization. However, one must make sure that the chosen inverse NN has enough layers and sufficient width for estimating the target structural heterogeneity one wishes to resolve—assuming that the observed data is not the limiting factor for the quality of the inversion.

It should be noted that enforcing both the free-surface constraint and the observed data from seismometers in the loss function (Equation 2) might seem redundant at first glance. Although the data collected by the seismometers implicitly impose a free-surface condition, our numerical simulations show that it is necessary to include a loss term for the free-surface constraint for an accurate recovery of the reflected phases at the free surface. The main reason is that in practical scenarios, seismometers are limited to only sparse spatial coordinates on the surface while information from the free-surface constraint is readily available through a differential equation, that is, $\nabla^2 \phi(x, t, z=0) = 0$ that can be evaluated at any position on the free surface.

2.5. Normalization

There are two important normalization steps in PINNs' implementation that are crucial to guarantee a convergence to the correct solution. First, the input and output variables of the network must be mapped to the interval $[-1, 1] \in \mathbb{R}$. Second, the acoustic wave PDE must be scaled such that both terms in Equation 1 are on the same order. See Appendix A for the appropriate formulations.

2.6. Optimization

Optimal values for the weights and biases of the proposed PINNs are obtained with the Adam optimizer (Kingma & Ba, 2014), an enhanced variant of the stochastic gradient descent with a learning rate of $1e-4$ and a suitably chosen batch size (BSGD) for each case study. We stop the gradient descent search when the improvement from the previous epoch (iteration) becomes negligible, or when we surpass a fixed number of iterations. We have tested our results with L-BFGS (Liu & Nocedal, 1989), a second-order optimizer that takes advantage of the Hessian matrix but did not observe significant improvement in the training.

Table 1 summarizes the batch size for each of the specific loss terms from Equation 2 for each case study. Note that unlike the PDE and the free-surface condition, we choose a fixed training data set to enforce the two early-time snapshots and the observed data from the seismograms. This is because the variability in the two latter data types is small enough to be captured with a relatively small data size. The optimal density of the different training data sets depends strongly on the ratio of the wavelength of the propagating wave to the domain size and the length scale of the structural heterogeneities as well as the total computational time.

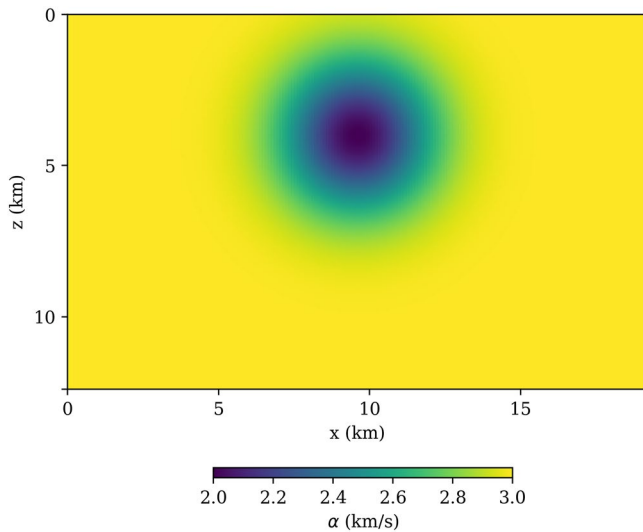


Figure 3. The distribution of the input wavespeed α for the forward problem.

2.7. Validation

We use numerical solutions obtained from SpecFem2D as the ground truth for the validation of the forward PINN model. We also use SpecFem2D to generate the early-time snapshot data and synthetic seismograms, so that we can test the PINN efficiency and accuracy. Any time-snapshots of the wavefield generated either analytically or with any numerical solver can be utilized and ultimately (in a future study) real data will replace the synthetically generated seismograms for the inversions. The SpecFem2D models are discretized on a 100×100 mesh spatially and we used a second-order explicit Newmark time stepping scheme with the time step-size $4e-5$ s. We employed perfectly matched layers (Komatitsch & Tromp, 2003) at the boundaries of the domain with a thickness of 10 nodes to simulate the absorbing boundary conditions for the point source cases, and the Stacey absorbing boundary conditions for the teleseismic plane wave source cases (Stacey, 1988).

3. Computational Experiments

In this section we examine the efficiency and accuracy of the PINN approach with the loss function given in (2). We present different case studies for forward and inverse modeling with the acoustic equation. We start with the forward simulation and then proceed to the inverse problems in the next section.

3.1. Case 1. Application of PINNs to Wave Propagation in a Heterogeneous Medium

Since our main focus in this paper is FWIs, we restrict our application of PINN forward acoustic wave propagation to a single case of heterogeneous medium. Figure 3 shows the wavespeed distribution in the domain with a background of 3 km/s perturbed with Gaussian heterogeneity of a minimum velocity of 2 km/s and a width (standard deviation) of 2.5 km. We choose a network with 4 hidden layers and 50 neurons per layer. The size of the training data set for the PDE loss term is 20,000 points while 3,600 points are set for each of the early-time snapshot data, picked from a normal distribution in space and time. We note that, for this particular case study, we do not impose a free-surface condition at the top of the domain, since our objective here is to observe how well PINNs can capture the wave propagation and interaction with the prescribed heterogeneity in the forward problem. The weights for different loss terms in Equation 2 are $\lambda_1 = 0.1$, $\lambda_2 = 1$, $\lambda_3 = \lambda_4 = 0$. We simulate 5 s of wave propagation. Figure 4 shows the evolution of the history of convergence of loss terms in Equation 2.

Figure 5 compares snapshots of the wavefield comparing PINN's prediction to the ground truth simulation using SpecFem2D. It is important to note that we are not using any labeled training data other than the two early-time snapshots of the wave propagation in the homogeneous regions of the domain, which serve to provide information about the acoustic source. In other words, the PDE is the sole information that the PINN uses to simulate the wave propagation in the heterogeneous part of the domain. Figure 5 shows that PINN provides a good solution to the wavefield. To avoid biases in network's parameters, all misfits are computed on a different input data set than the ones used for training.

From Figure 5 we observe that PINNs handle absorbing boundary conditions effectively. Absorbing boundary conditions can be notoriously challenging to implement accurately. We find that PINNs enforces absorbing boundaries by default if no boundary condition is mathematically prescribed in the training (see misfit in Equation 2).

3.1.1. Application of PINNs to Inverse Modeling

In this section, we present the application of PINNs to FWIs. For all case scenarios, we follow the same strategy outlined here. We build a 2-D domain with prescribed wavespeed distribution and simulate the acoustic wave propagation using SpecFem2D. We then use the generated seismograms as the data (observation) to run the inversion. Again, we use two early-time snapshots of the displacement field from SpecFem2D as the training data.

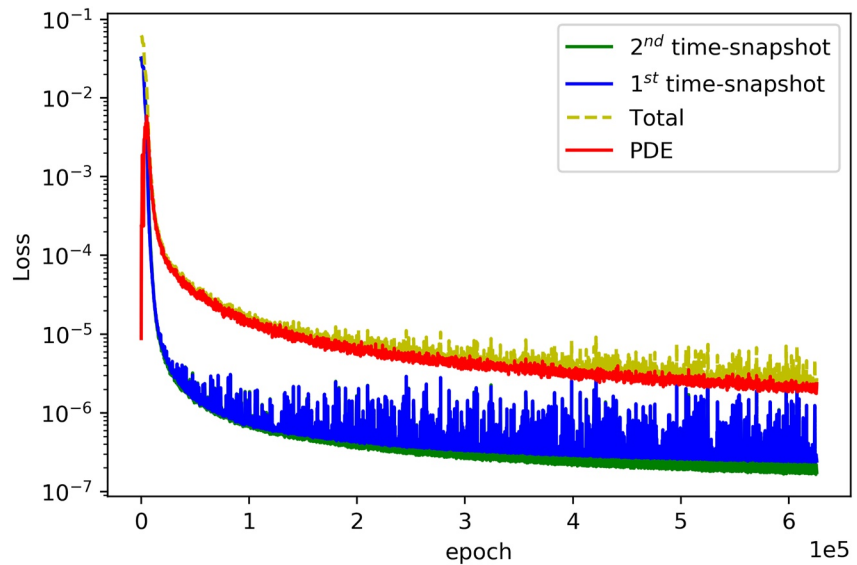


Figure 4. Evolution of the different loss terms in Equation 2 as well as the total loss for the forward problem.

It is important to note that we only use snapshots taken before the wave interacts with any heterogeneities in the ground truth model, so as to avoid providing more information into PINN than would normally be available in practical inverse problems encountered in seismology. In all but the last example, we impose a free-surface condition at the top of the domain. We test the PINN's performance on recovering discontinuous as well as continuous velocity anomalies with various seismic source types such as a single point source, collection of point sources and teleseismic plane waves. We also assess the PINN's ability to capture free-surface reflections and absorbing boundary conditions in these case studies.

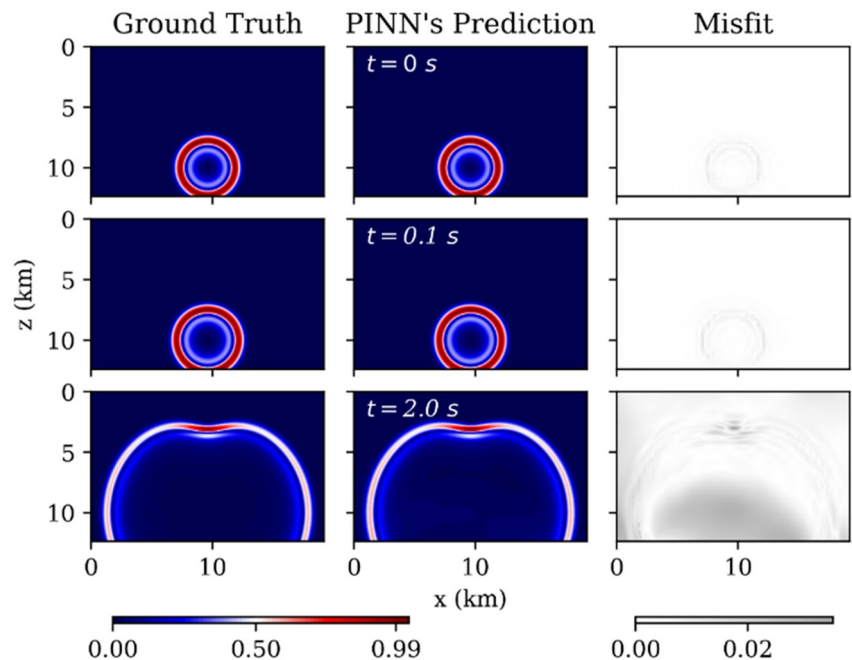


Figure 5. Ground truth versus predicted magnitude of the wavefields from physics-informed neural networks for the forward problem. The first two wavefield snapshots from SpecFem2D at $t = 0$, $t = 0.1$ s are used as the training data. The misfits show the absolute pointwise error.

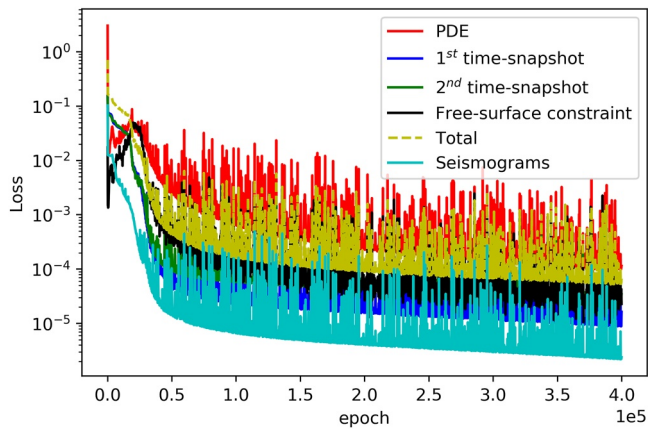


Figure 6. Evolution of the different loss terms in Equation 2 as well as the total loss for the synthetic crosswell experiment in a homogeneous domain.

3.2. Case 2. Crosswell Experiment: Homogeneous Velocity Model With a Single Point Source

We start with the simplest case study, a homogeneous domain with wavespeed 3 km/s. We use PINNs to retrieve the wavespeed across the domain in a synthetic crosswell experiment given the set of training data outlined above. We choose a network with 4 hidden layers and 50 neurons in each. The size of the training data set for the PDE loss is 10,000, each of the early wavefield time-snapshots adds 3,600 data points, and the free-surface constraint finally adds another 5,000, all chosen from normal distributions in space and time. We use data points from a 0.4 s time window with a sampling frequency of 200 Hz from each component of the seismograms' time series and from 20 equally distanced seismometers (total 3,240 data points from all seismometers) with depths ranging from the top of the model and extending to 450 m (Figure 7a). The seismic wavefield is generated from one point-source with a Gaussian source time function and a dominant frequency of 20 Hz located on the left side of the domain (Figure 8). The weight values for different loss terms in Equation 2 are $\lambda_1 = \lambda_3 = 0.1$, $\lambda_2 = \lambda_4 = 1$. Figure 6 shows the evolution of the different loss terms until convergence.

Despite the relatively poor initial guess for the wavespeed, we can see that PINN successfully recovers an accurate estimate of the domain's wavespeed (Figure 7) as well as the wavefield (Figure 8). The quality of the solution is further supported by the great match between the synthetic (observation) and calculated seismograms (Figure 9).

At the top surface, where we imposed free-surface constraints, we observe a good match between the reflected wave simulated with SpecFem2D applied to the true wavespeed model and the PINN solution in terms of amplitude, waveform, and timing (Figure 8).

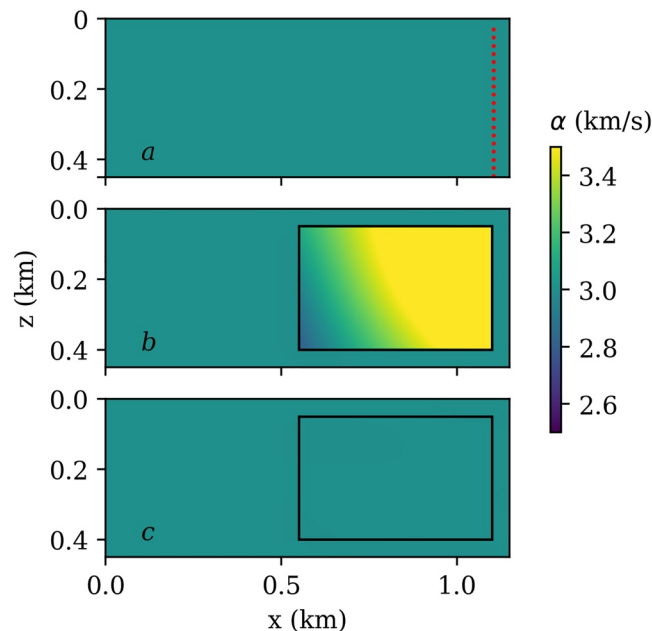


Figure 7. (a) True domain, (b) Initial guess (c) physics-informed neural network's (PINN's) inversion results after convergence. The black rectangles in panels (b) and (c) show the area where the inversion is performed with PINN. The red dots in panel (a) show the locations of the seismometers.

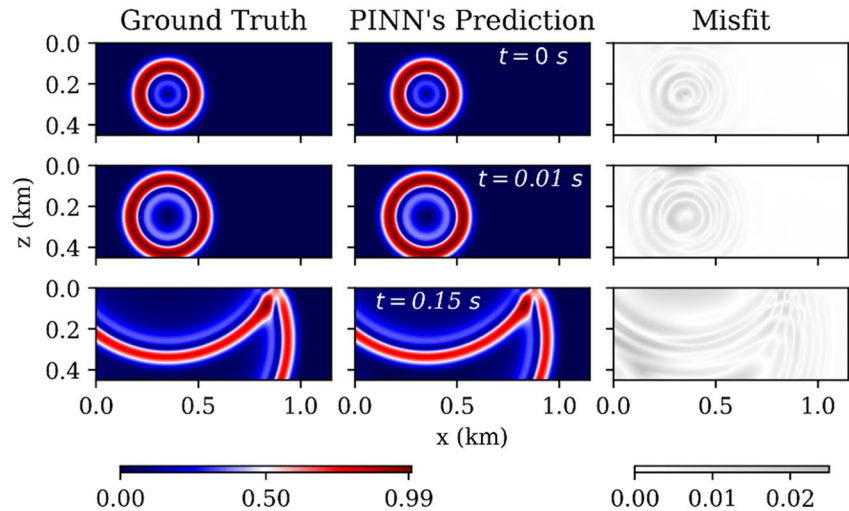


Figure 8. Comparison between ground truth and modeled wavefields and their absolute pointwise differences for the synthetic crosswell experiment with a homogeneous wavespeed distribution. The first two wavefield snapshots from SpecFem2D at $t = 0$, $t = 0.01$ s are used as training data.

It is important to emphasize again that we have not explicitly enforced absorbing conditions in the loss function. The ability of PINN to manage automatically absorbing conditions stem from PINNs solutions being smooth and infinitely differentiable functions. Both components of the time series of the synthetic seismograms (observations) are also matched accurately by the PINNs solution (Figure 9).

3.3. Case 3. Crosswell Experiment: Ellipsoidal Velocity Anomaly With a Single Point Source

This case study is designed to test the ability of PINNs to recover sharp 2-D anomalies from a synthetic crosswell experiment. An ellipsoidal low velocity anomaly with wavespeed of 2 km/s is embedded in a homogeneous background model (3 km/s). The velocity contrast between the anomaly and background is sharp (step function). The size of the domain, the location of the source and seismometers and the sampling frequency of the training data are the same as the previous case (Figure 10a). We design a network with 8 hidden layers and 100 neurons per layer for the wavefield NN. The size of the training data set for the PDE loss is increased to 40,000 points and remains 3,600 and 5,000 for each of the early-time snapshot data and the free-surface constraint, respectively, again chosen from a normal distribution in space and time. Furthermore, we set the weights of the misfit function $\lambda_1 = \lambda_3 = 0.1$, $\lambda_2 = \lambda_4 = 1$.

Figure 10 shows the inverted solution for the acoustic wavespeed in comparison to the ground truth and the starting model. PINN successfully retrieves the location, dimension, and magnitude of the anomaly. The inverted solution is smoothed instead of the sharp discontinuous transition in material property of the true model.

A comparison of the wavefield after 0.15 s is provided in Figure 11. It shows that the inverted solution matches the wavefield of the synthetic true solution accurately. The match between observed and modeled seismograms for each of the 20 seismometers is also excellent (Figure 12). For some of the seismograms, the slightly larger discrepancies between the input data and the outputs from PINN around the end of the time window analyzed is a regression artifact independent of the method. There, the optimization close to the final time is not perfect. Nevertheless, the prominent parts of the time series have been closely matched by PINNs solution and increasing the sampling frequency would improve the remaining misfit.

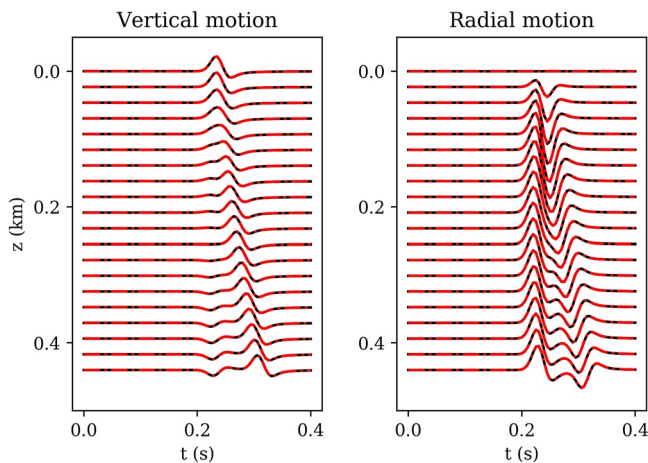


Figure 9. The comparison between ground truth (black line) and physics-informed neural networks's prediction (red dashed line) of the vertical and radial component seismograms for the synthetic crosswell experiment with a homogeneous wavespeed distribution. Locations of the seismometers are shown in Figure 7a.

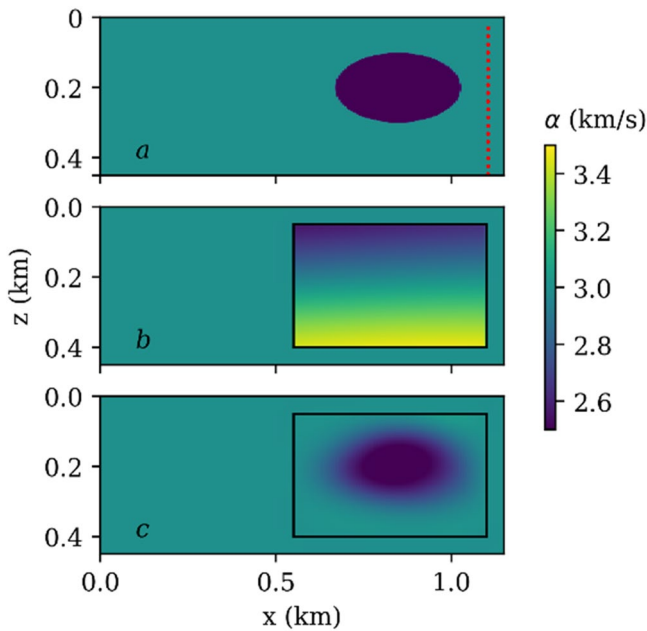


Figure 10. (a) True domain, (b) Initial guess (c) physics-informed neural network's (PINN's) inversion results after convergence. The black rectangles in panels (b) and (c) show the area where the inversion is performed with PINN. The red dots in panel (a) show the locations of the seismometers. Note in the true domain the wavespeed transition from the ellipsoidal anomaly to the background is discontinuous.

We have also performed this inversion with a smaller NN with 4 hidden layers and 50 neurons per layer. The smaller network still yields a good estimate of the wavespeed distribution, however the recovered wavefield is less accurate and justifies the choice of a deeper network for an improved accuracy (See Supporting Information S1).

3.4. Case 4. Ellipsoidal Anomaly With Two Incident Plane Waves

In teleseismic imaging, plane waves from multiple incidence angles are generally required for a suitable ray coverage of the area under study. In the next example, we demonstrate how to incorporate multiple events into PINNs' formulation in a consistent manner such that it avoids substantial increase in computational costs. Exploiting the linearity of the acoustic wave equations, we superpose multiple events and design a network to simulate the resulting superposition of all events at once. This approach avoids defining a network for each event, which can become prohibitively expensive.

In this case study, we generate two compressional plane waves with a ricker source time function and a dominant frequency of 2 Hz with incidence angles $+20^\circ$ and -20° (with respect to the vertical axis). The two plane waves propagate through an ellipsoidal low velocity anomaly (wavespeed of 2 km/s) embedded in a homogeneous background model with 3 km/s. We set up 17 equally spaced seismometers at the top surface of the model starting from the top leftmost corner of the domain and extending 15 km to the right (Figure 13a). We record 5 s of the seismic signal with a sampling frequency of 20 Hz for training the PINN (3,400 training data from the seismometers' time series). A PINN with 8 hidden layers and 100 neurons

per layer is used for the wavefield NN. The training data sets for the loss terms corresponding to PDE, free-surface constraint on top and each of the wavefield early time-snapshots consist of 40,000, 5,000 and 3,600 data points, chosen from a normal distribution in space and time, respectively. Furthermore, we set the weights of the misfit function $\lambda_1 = 0.01$, $\lambda_2 = \lambda_3 = \lambda_4 = 1$. Figure 13 demonstrates the excellent agreement in terms of location, dimensions, and magnitude of the anomaly between the ground truth and PINN's inversion results.

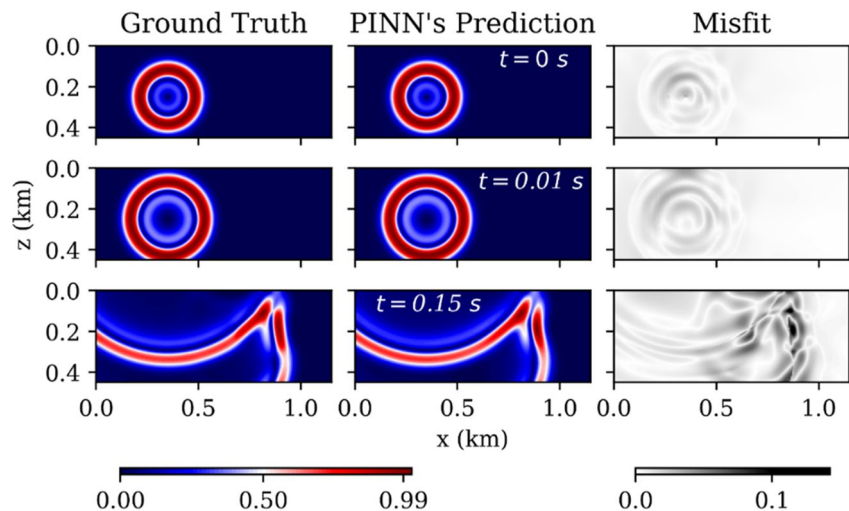


Figure 11. Comparison between ground truth and modeled wavefields and their absolute pointwise differences for the synthetic crosswell experiment with a discontinuous ellipsoidal anomaly. The first two wavefield snapshots from SpecFem2D at $t = 0$, $t = 0.01$ s are used as the training data.

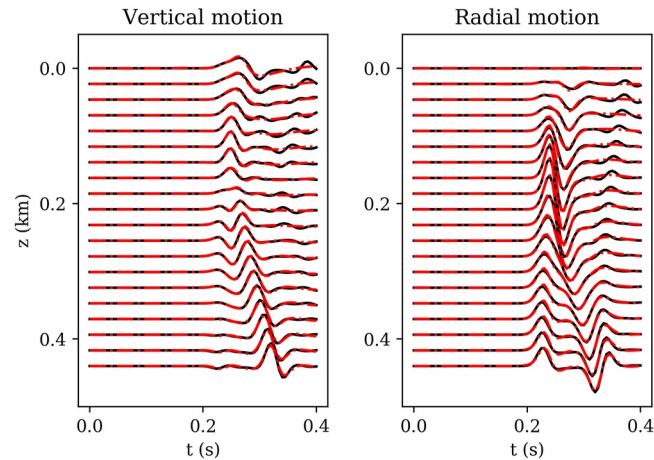


Figure 12. The comparison between ground truth (black line) and physics-informed neural networks's prediction (red dashed line) of the vertical and radial component seismograms for the synthetic crosswell experiment with a discontinuous ellipsoidal anomaly. Locations of the seismometers are shown in Figure 10a.

The amplitude and structure of the recovered wavefield from PINN follows closely the forward simulations obtained from the numerical solver (Figure 14). However, we observe a slight timing difference at different parts of the wavefield at later times (Figure 14, lower most right panel). This time-difference is due to the fact that PINN recovers a smooth version of the structural heterogeneity (as in any other inverse technique), and since it

solves the forward and inverse problems in a coupled fashion, the forward solution (wavefield) takes effect from the smoothed inverted structure. The smoothing associated with the inversion is also responsible for the absence of certain phases reflected off the anomaly. On the other hand, PINN is able to capture the reflected waves off the free surface on top (Figure 14). This shows that when the interface condition is explicitly enforced, PINN can capture reflections. Moreover, we observe the PINNs' great ability in simulating absorbing boundary layers at the left, right and bottom edges of the computational domain, without explicitly enforcing them (Figure 14). Our results show that the bulk of the training is spent on improving small discrepancies, while PINN converges to a satisfying velocity model in less than 70,000 epochs, instead of the 400,000 used for a complete training in this case (See Supporting Information S1).

From Figure 15, we observe that PINN finds a precise fit to the observed synthetic seismograms ensuring a correct capture of the physics of the wave's interaction with different parts of the domain. For instance, there is a sharp deficit in the recorded energy from the observed vertical component seismograms at around 9 km, signaling a “shadow” region affected by a low velocity anomaly at depth, which is impressively captured by PINNs as well. On the other hand, the absolute zero energy on the radial component seismograms, signature of a free-surface physical constraint in acoustic media is also correctly retrieved by PINN.

3.5. Case 5. Checkerboard Test With Nine Point-Sources

For the last case study, we test PINNs' performance when inverting for a sinusoidal checkerboard model velocity with alternating positive and negative anomalies, as depicted in Figure 16a. We also aim to see if a reasonably sized NN can capture the complex response of the system to as many as nine point-sources. The point locations for 20 seismometers are set at the top

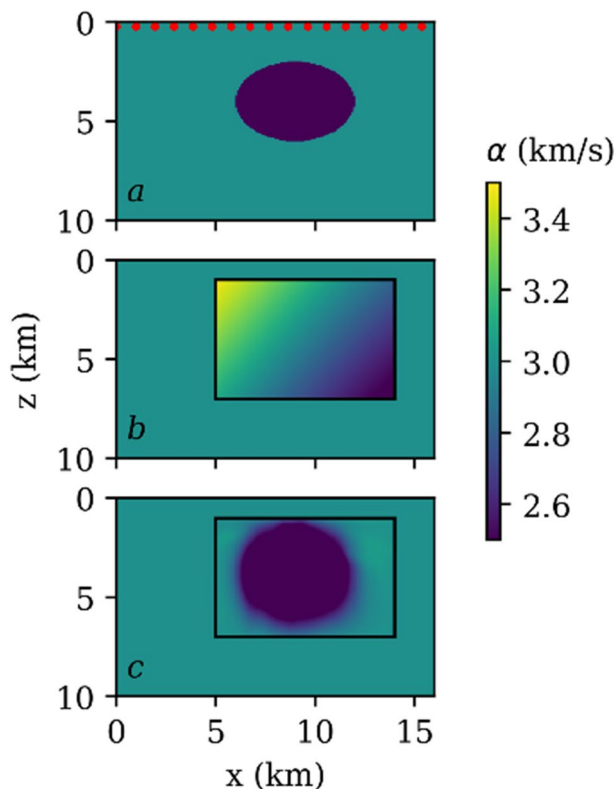


Figure 13. (a) True domain, (b) Initial guess (c) physics-informed neural network's (PINN's) inversion results after convergence for the teleseismic case study. The black rectangles in panels (b and c) show the area where the inversion is performed with PINN. The red dots in panel (a) show the locations of the seismometers.

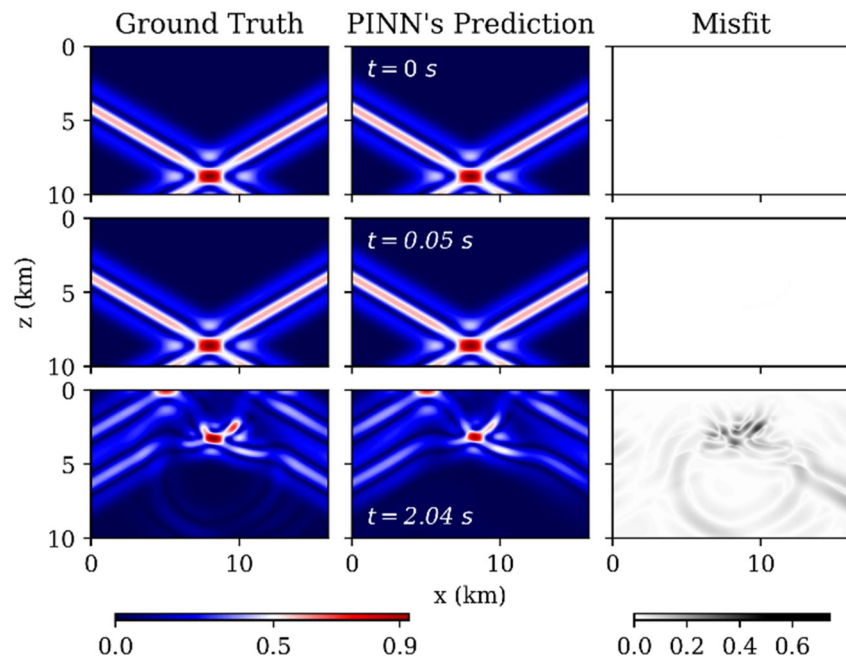


Figure 14. Comparison between ground truth and modeled wavefields and their absolute pointwise differences. The first two snapshots from SpecFem2D at $t = 0$, $t = 0.05$ s are used as the training data.

surface at equal distance from each other, starting at 2.6 km from the left side of the domain extending to 17.6 km to the right. We use nine point-sources all with a Gaussian source time function and a dominant frequency of 2 Hz (Figure 17). We use a NN with 10 hidden layers and 100 neurons per layer. The size of the training data sets for the PDE, and each of the early-time snapshots are 60,000 and 3,600, respectively. We also use data from the seismometers with a sampling frequency of 50 Hz (11,600 total data points from seismometers' time series). The different loss term weights are set $\lambda_1 = 0.1$, $\lambda_2 = \lambda_3 = 1$, $\lambda_4 = 0$. For this case we do not impose a free-surface constraint at the top of the domain. From Figures 16 and 17 we observe that PINN is capable of recovering a complex oscillatory velocity gradient and the corresponding wavefield. The synthetic seismograms from SpecFem2D have also been successfully matched by PINN's outputs with negligible misfit (Figure 18).

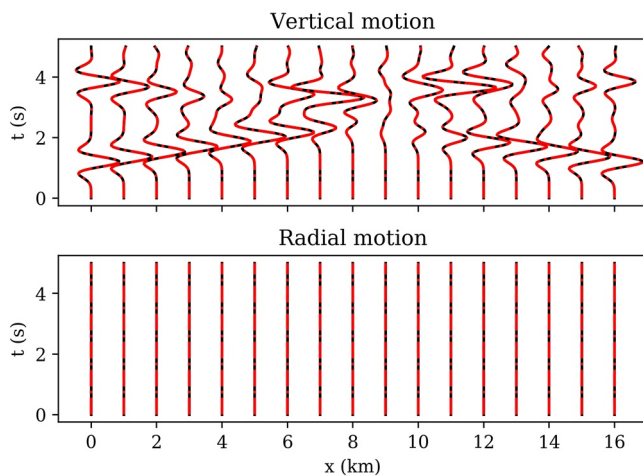


Figure 15. The comparison between ground truth (black line) and physics-informed neural networks' prediction (red dashed line) of the vertical and radial component seismograms for the teleseismic case study. Locations of the seismometers are shown in Figure 13a.

4. Discussion

We have shown that PINNs are able to accurately solve the forward and inverse modeling of the acoustic wave equation in complex media. PINNs are meshless and have impressive generalization capability and given only a set of two early-time snapshots they can predict the wavefield solution much later in time. Also, with minimal observed data only at the surface, and without any training data within the computational domain (other than the early-time snapshots) PINNs yield excellent results for seismic inversions. This shows that the PINN formalism can be implemented independent of other numerical solvers that require more complex implementation.

We extended the original approach by Raissi et al., 2019 by incorporating multiple seismic events into one NN and hence greatly optimizing the inverse problem for practical applications. Different seismic sources (plane waves and point sources) with variable frequencies can be implemented with limited effort. However, it must be noted that if the energy from the superposition of seismic sources interferes destructively at the site of a

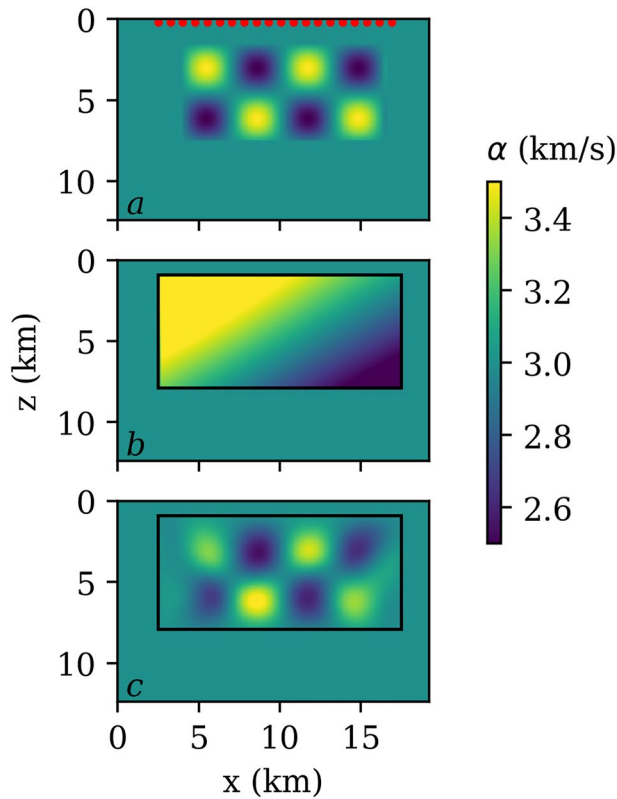


Figure 16. (a) True domain, (b) Initial guess (c) physics-informed neural network's (PINN's) inversion results after convergence. The black rectangles in panels (b and c) show the area where the inversion is performed with PINN. The red dots in panel (a) show the locations of the seismometers.

seismometer, superposition must be avoided because of the loss of information and the consequent deterioration of the solution. In this case, a possible approach is to define different groups of non-destructively interfering events and then define an independent NN for each group. This approach will still decrease the computational costs compared to defining a NN for each event.

Our study shows that the computational costs (size of the NNs and number of epochs to reach convergence) of the problem is strongly influenced by the frequency of the source and the complexity of the wavespeed structures. We observe that a smaller NN is capable of yielding satisfactory results for an inverse problem if the structure of the medium is simple. For instance, the NN for the homogenous inversion model (Case 2) is significantly smaller than the NN for the model with an ellipsoidal anomaly (Case 3). The NN's setup is also influenced by the frequency content of the source. Modeling higher frequencies (for the same physical dimensions) requires a larger training data set for the PDE loss to capture finer features such as refracted waves which are more pronounced at higher frequencies.

PINNs are a meshless method. They offer great flexibility in terms of implementation and if a priori knowledge is available for inverse problems. Moreover, our results show that PINNs perform well even without a priori knowledge or in the absence of an educated starting model. Nevertheless, one can easily implement a priori constraints on the wavespeed distribution. For instance, if we expect a medium with dominantly vertical variations in acoustic/seismic velocities, it is straightforward to set the functional form of the inverse NN to explicitly depend only on one spatial variable, namely z . This can be achieved with no fundamental change to the original algorithm.

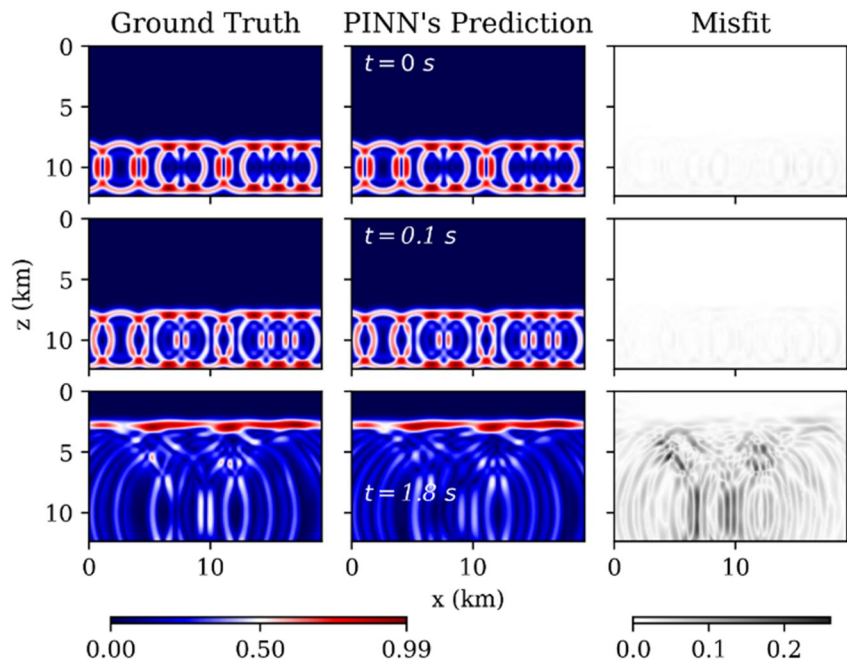


Figure 17. Comparison between ground truth and modeled wavefields and their absolute pointwise differences. The first two snapshots from SpecFem2D at $t = 0$, $t = 0.1$ s are used as the training data.

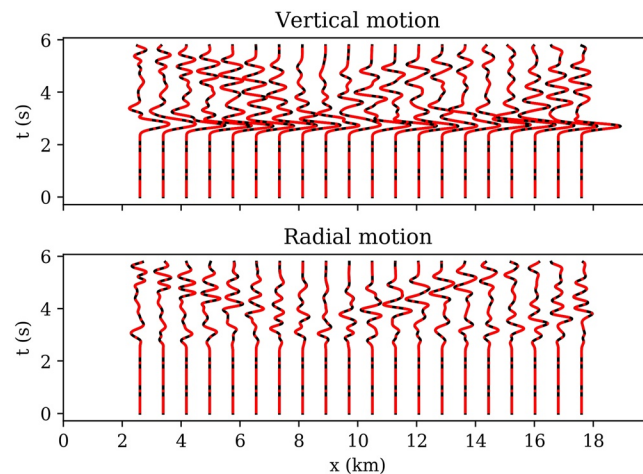


Figure 18. The comparison between ground truth (black line) and physics-informed neural networks's prediction (red dashed line) of the vertical and radial component seismograms for the checkerboard case study. Locations of the seismometers are shown in Figure 16a.

PINN's implementation is made simple and compact thanks to modern python's libraries such as TensorFlow and PyTorch. For example, the script (available on <https://doi.org/10.26300/x3wd-4k56>) utilizes only a few hundred lines to run both the wave propagation and optimization for the inversion. It is also easy to modify or change the constraints imposed at the boundaries of the domain as they are expressed only in the definition of one of the loss terms in Equation 2.

For the case studies with a discontinuous true model, reducing the smoothening effect of the inversion is possible if one uses a significantly larger training data set for the PDE part of the loss function and includes a priori knowledge of the presence of a discontinuity or a supervised distribution of many training data around the discontinuity. Moreover, a different type of minimization norm, for example, L1, may lead to sharper discontinuities.

A challenge when using PINN comes from the heuristics nature of the algorithm when selecting the proper weights for each loss terms or selecting the proper network size. Work remains necessary to better constrain the factors that control the convergence of PINNs and the criteria for choosing the minimum density of the training data for the PDE to guarantee convergence to the correct solution (Shin et al., 2020; B. Wang et al., 2020; S. Wang, Teng, & Perdikaris, 2020; S. Wang, Yu, & Perdikaris, 2020); see also the recent work of McClenny and Braga-Neto (2020). Moreover, for large computational domains, efficient strategies to implement PINNs on GPUs are required to deal with the large memory cost involved. The implementation of PINNs with a domain decomposition in space and space-time domain is successfully studied by Jagtap et al. (2020) and Jagtap and Karniadakis (2020), which is further extended to a multi-GPU platform by Shukla, Jagtap, & Blackshire (2021), Shukla, Jagtap, and Karniadakis (2021). Therefore, the computational efficiency of proposed method for larger and elastic approximation can be successfully tamed.

4.1. Present Challenges and Future Work With PINNs and FWIs

This study aims to present a proof-of-concept that PINNs can be applied to acoustic wave forward and inverse modeling. In this contribution we have mostly relied on developing a framework based on the original formulation of PINNs introduced by Raissi et al., 2019. The different case studies proposed above demonstrate the applicability of PINN to acoustic waveform modeling and some unexpected benefits with regards to its versatility with boundary conditions. However, as we show in the present section, some challenges remain when applying the same strategy to more complex structures such as the Marmousi model for example, (Martin et al., 2002; Versteeg, 1994).

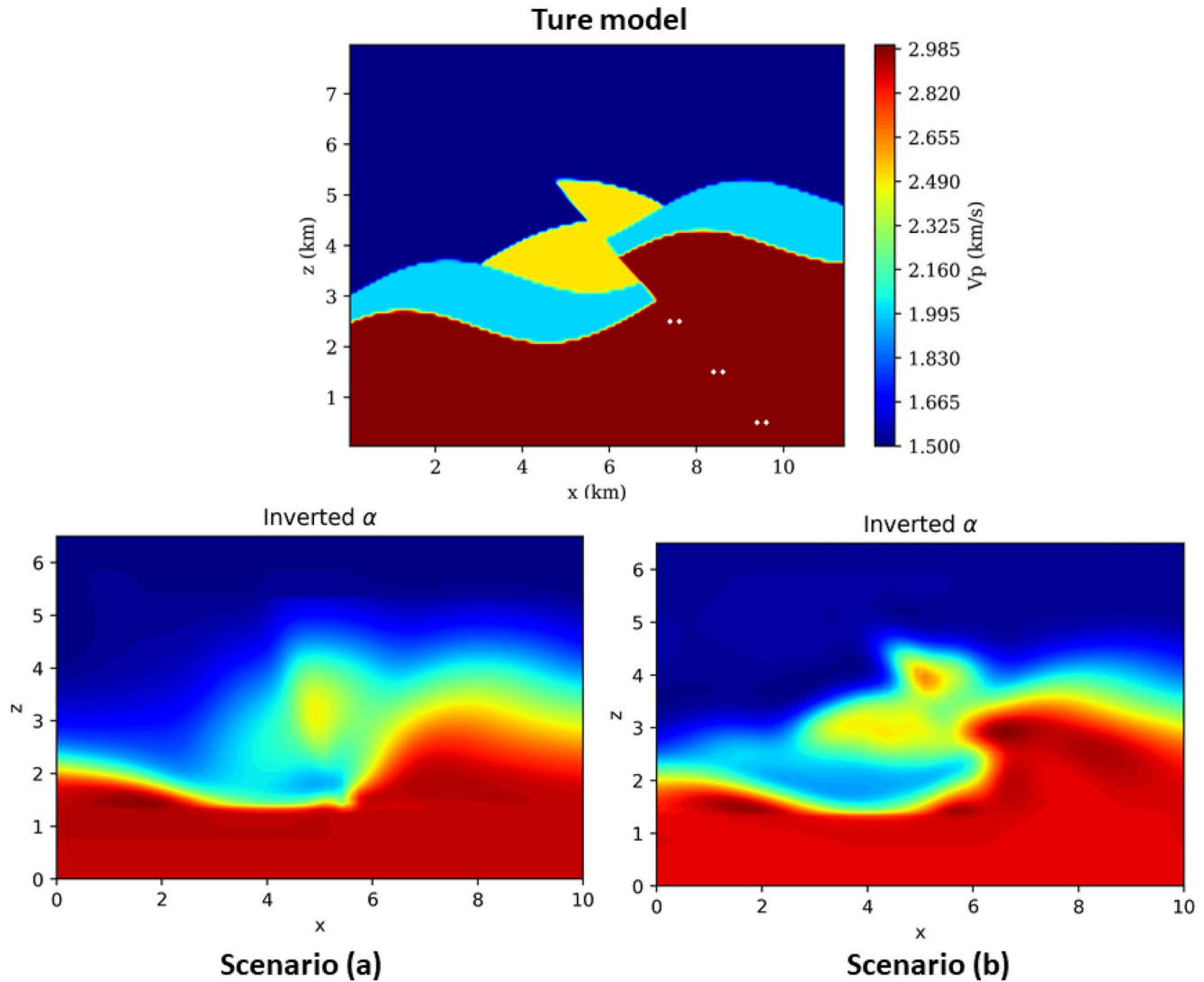


Figure 19. True model (top) and two inversion results for each of the scenarios described in the main text with different amount of training data given in Figure 20.

As an example, we perform an inversion for a synthetic model with sharp gradients in velocity shown in Figure 19. We consider two inversion scenarios:

1. First, we perform an inversion using the same sets of data as in cases presented earlier in the manuscript, that is, using two early snapshots of the wave propagation in addition to the full time-series data from seismometers at the surface and the PDE training data.
2. In the second approach we add three more snapshots of the wave propagation at later stages of propagation obtained with SpecFem2D (see Figure 20). We note that this is not a realistic scenario as snapshots of the wave propagation in the true structure are not available in real case applications. However, this is an important case study to demonstrate how the addition of the new data improves the inversion results significantly.

We use 10 point-sources (5 Hz) at the bottom of the computational model and 21 seismometers equally spaced at the surface. The inversion results for these two case studies are shown in Figure 19. For both case studies we use a NN with 6 hidden layers and 100 neurons in each layer and trained for the same duration of time (same number of optimization epochs) to the final convergence stage. We observe that the excessive smoothing observed for scenario (a) is not an intrinsic limitation of PINN, given that the addition of ulterior snapshots that provide more information about the structure allows us to retrieve a significantly improved inverted model.

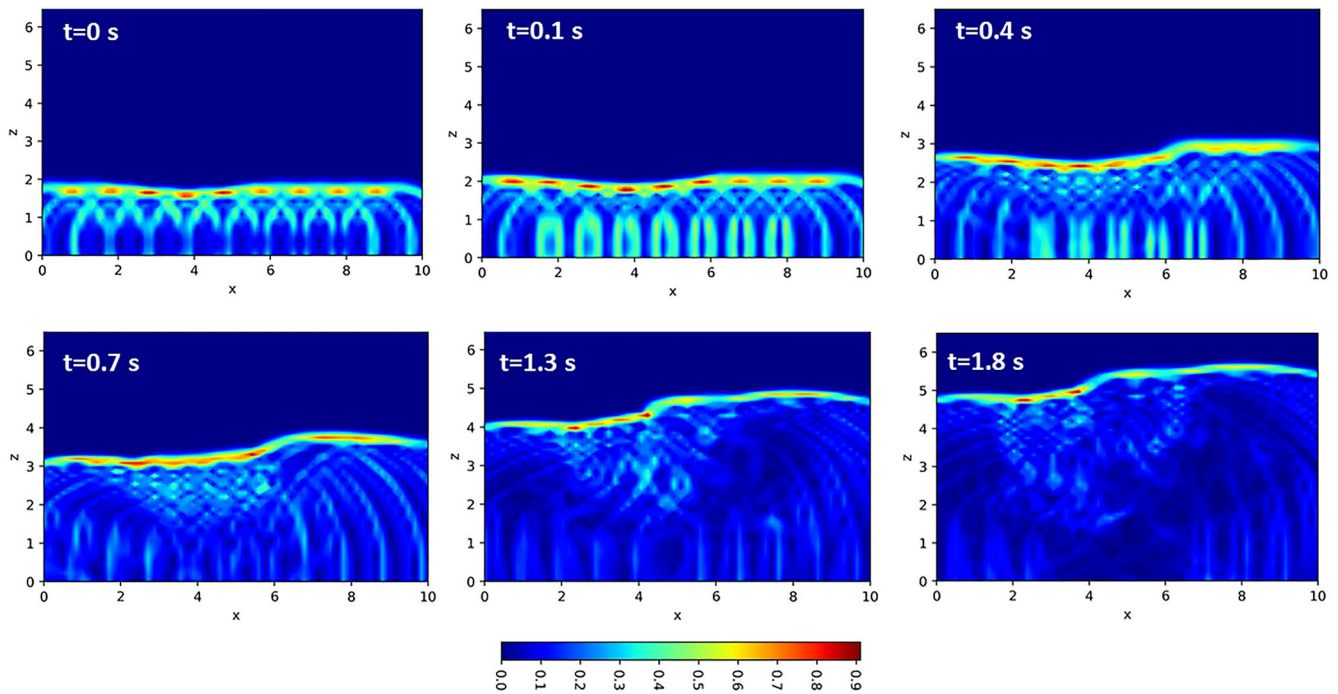


Figure 20. Snapshots of the training wavefield (5 Hz) used for inversion case studies of an Earth model with fault-type features. For the scenario (a) in Figure 19 we use only two snapshots at times $t = 0$ s, $t = 0.1$ s but for the case study (b) we use all six snapshots.

This suggests that the challenges presented by more complex wavefields do not reveal an intrinsic limitation of PINNs.

Moving forward to expand PINNs to more complex and realistic Earth models, we argue that leveraging novel/emerging developments in field of PINNs will be key to capture more effectively the wavefields that probe more complex structures. As an example, Figure 21 illustrates the complex wavefield generated from a single point source in the Marmousi model from a simulation with SpecFem2D. As discussed throughout the manuscript, the current implementation of the model struggles to identify all the internally reflected phases at sharp material discontinuities. This current limitation may be caused by the current formulation of the PINN

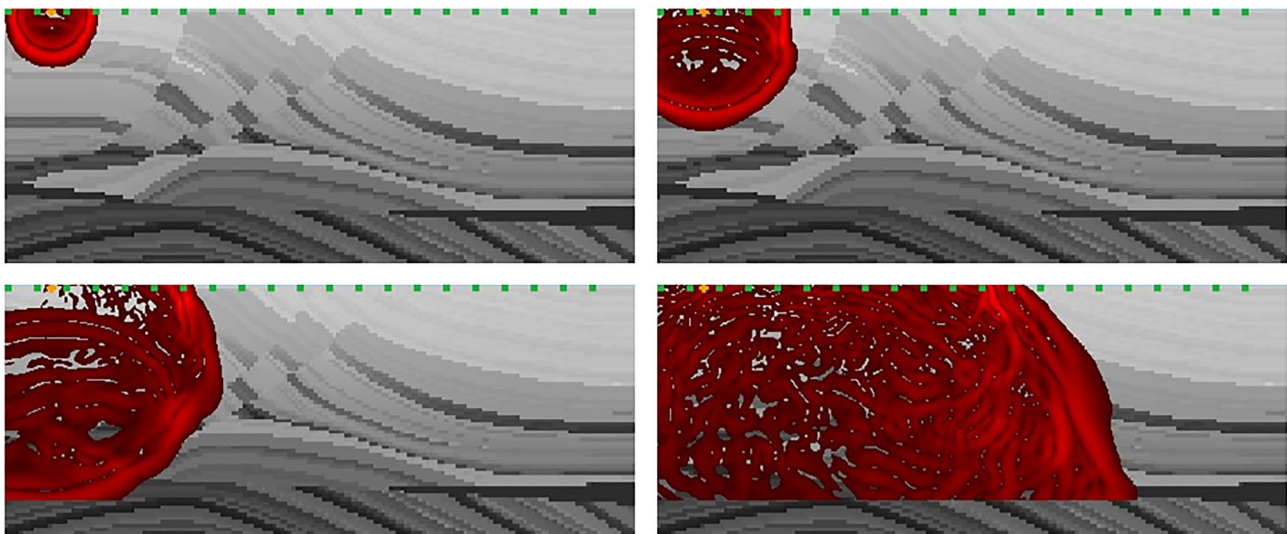


Figure 21. Snapshots of the propagation of a 10 Hz acoustic point source in a Marmousi model simulated with Spectral Element Method (SpecFem2D).

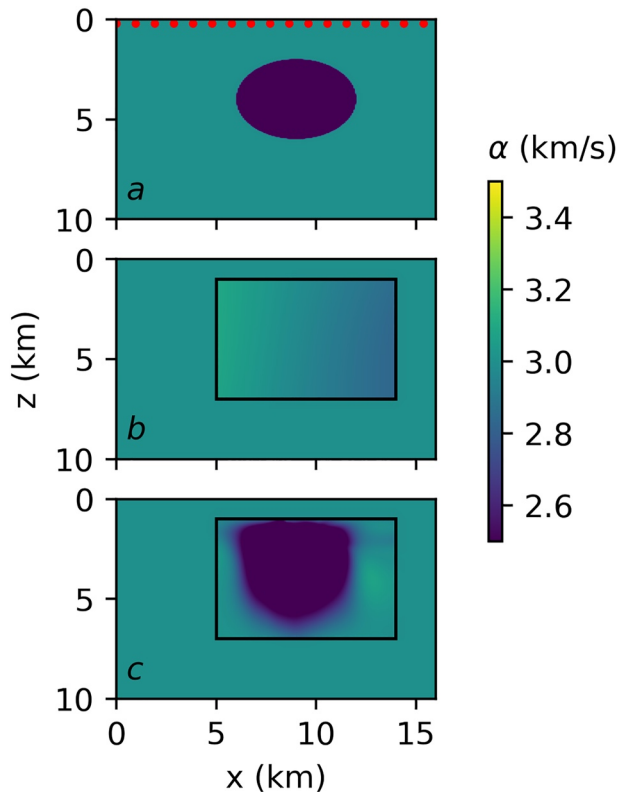


Figure 22. (a) True domain, (b) Initial guess (c) physics-informed neural network's (PINN's) inversion results after convergence (setup here is similar to Case 4). The black rectangles in panels (b and c) show the area where the inversion is performed with PINN. The red dots in panel (a) highlight the location of the seismometers.

model. Recent developments in the field of physics-based machine learning offer compelling alternatives to tackle this issue. For instance, conservative PINN (cPINN; Jagtap et al., 2020) and extended PINN (XPINN; Jagtap & Karniadakis, 2020) provide means to decompose the computational domain into multiple subdomains and define a separate NN in each subdomain. cPINN and XPINN will potentially enhance the flexibility of the method and improve its efficiency when dealing with complex wavefields. B. Wang et al., 2020; S. Wang, Teng, & Perdikaris, 2020; S. Wang, Yu, & Perdikaris, 2020; W. Cai et al., 2020 and Jagtap et al., 2020a, 2020b also propose new treatments to the activation function and network architecture to avoid the well-known spectral bias of the neural networks, where the network cannot learn the high frequency components of the solution in a reasonable time frame. Alternatively, we suggest that a hybrid method could potentially be beneficial whereby intermittent calculations from a conventional forward solver informs PINNs of the complex scattering pattern during the training steps.

5. Conclusion

We have demonstrated that PINNs are able to solve the wave propagation and FWIs by relying on information from governing partial differential laws when there is limited data available. We have shown how inversions with multiple events can be performed with PINNs with limited additional memory or computational cost. PINNs' seamless ability to handle diverse sets of constraints, their meshless nature and the simplicity in formulation/implementation open new perspectives as the next generation of inverse solvers for geophysical applications. They provide a flexible framework to incorporate multiple data types and/or any a priori knowledge of the structure imaged and hence offer a great potential for joint and Bayesian inversions (Yang et al., 2021).

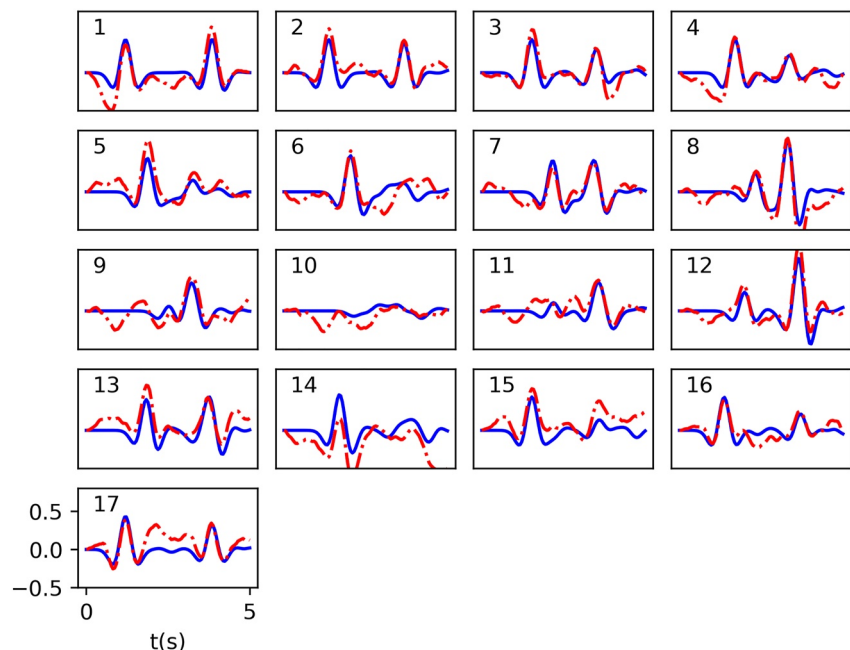


Figure 23. Raw signal (blue continuous line) and the signal contaminated with white noise (red dashed line) used for inversion. The number in each panel shows the location of the seismometers shown on Figure 22 from left to right.

Appendix A

Here we show the two important normalizations used in Physics-Informed Neural Network's (PINN's) formulation in this study:

1. We map each input variable to the NN onto the interval $[-1, 1] \in \mathbb{R}$ (Raissi et al., 2019):

$$X \rightarrow \frac{2X}{\max(X)} - 1$$

2. We scale the wave PDE in Equation 1 to constrain the wavespeed in the $[0, 1] \in \mathbb{R}$ interval. Using the scaled spatial coordinates $x = \max(\alpha)x^*$, $z = \max(\alpha)z^*$, yields,

$$\alpha^{*2} \nabla^{*2} \phi = \frac{\partial^2 \phi}{\partial t^2}$$

where $\alpha^* = \alpha / \max(\alpha)$ and $\nabla^{*2} \equiv \frac{\partial^2}{\partial x^{*2}} + \frac{\partial^2}{\partial z^{*2}}$.

Appendix B

In this section we explore PINNs ability to handle seismic data contaminated with white noise. We use case study (4) and add noise to the synthetic data before performing the inversion. We use a Gaussian white noise with a signal-to-noise ratio of 1.16 (energy of the signal divided by the energy of the noise) and filter the noisy signal in the 2–6 Hz frequency range using a butterworth filter (Figure 23). The setup of the problem, the NN's parameters and training data size, are the same as in case study (4).

Despite the presence of noise, Figure 22 shows a good agreement between the ground truth and PINN's inverted model in terms of location, dimensions, and velocity contrast of the anomaly. This result suggests that the training on the governing equations in PINN is likely to provide a robust framework to filter out noise and perform a satisfying inversion.

Free-surface constraint:	Stress-free condition imposed at the top of the model to simulate the surface of the Earth.
Boundary data:	Any data for the wavefield located at the boundaries of the computational domain. This can be either in the form of observed data (seismograms), or in the form of imposed conditions such as a free-surface or absorbing conditions.
Early-time snapshots:	Data from the snapshots of the wavefield recorded at $t = 0$ and $t = \Delta t$ obtained here with a wave equation solver (SpecFem2D) that are used as inputs to the PINN to train for characteristics of the seismic source in space and time.

Data Availability Statement

A sample script in python for this study can be found on Brown University's data repository: <https://doi.org/10.26300/x3wd-4k56>.

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